CASCADIA ANALYTICS

Alphabet Inc.

Equity Research Note | GOOGL | 22 OCTOBER 2026

The business is exceptional but the price discounts perfection

Alphabet Inc. screens as a Hold, Valuation Concerns with a $160.95 blended target, 50% below the current price.

Analyst: Justin Wen

REFERENCE PRICE	BLENDED TARGET	BLENDED UPSIDE	
$322.48	$160.95	-50%	HOLD, VALUATION CONCERNS

Investment Thesis

The business is exceptional but the price discounts perfection. Alphabet owns the two most durable franchises in technology, Search and YouTube, plus the fastest-scaling cloud business in the industry. Our framework rates the business Attractive on quality, growth, and trend, yet the DCF puts fair value far below the market price. This is a great company at a demanding price. The capex supercycle is the crux of the valuation debate. Management guided 2026 capex to $195-205B and signaled a significant increase

- The business is exceptional but the price discounts perfection
- The capex supercycle is the crux of the valuation debate
- Cloud is the growth engine, Search is the cash engine under watch
- Valuation verdict: Hold, Valuation Concerns

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The latest print

The capex supercycle is the crux of the valuation debate. Management guided 2026 capex to $195-205B and signaled a significant increase again in 2027. Q2 free cash flow turned negative for the first time as $44.9B of quarterly capex exceeded $39.1B of operating cash flow. The bull case requires this spend to convert into durable Cloud and AI revenue at high incremental margins; the bear case is a multi-year ROI shortfall.

Cloud is the growth engine, Search is the cash engine under watch. Google Cloud revenue grew 82% YoY to $24.8B with operating income tripling to $8.8B and a $514B contracted backlog. Search keeps compounding, but Meta's Muse agent (2.8M downloads in twelve days) and court-ordered data-sharing remedies are the first credible threats to the ad moat in two decades.

Valuation verdict: Hold, Valuation Concerns. Our probability-blended DCF target is $160.95, implying 50% downside from $322.48. Per our rating rules, an overvalued but healthy business is a Hold on valuation concerns, not a Sell. We would turn constructive on a capex normalization signal, a Search monetization re-acceleration, or a materially lower entry price.

$120.0B	34.0%	$24.8B	$514B	$195-205B	-$5.9B
Q2 2026 REVENUE	Q2 2026 OPERATING MARGIN	GOOGLE CLOUD Q2 REVENUE	CLOUD BACKLOG	2026 CAPEX GUIDANCE	Q2 2026 FREE CASH FLOW

Key metrics in detail

METRIC	LATEST	VS YEAR AGO	TREND
Q2 2026 Revenue	$120.0B	+24.4% YoY	Accelerating
Q2 2026 Operating Margin	34.0%	+1.9pp	Expanding
Google Cloud Q2 Revenue	$24.8B	+82% YoY	Hypergrowth
Cloud Backlog	$514B	~2x	Surging
2026 Capex Guidance	$195-205B	+119% vs 2025	Supercycle
Q2 2026 Free Cash Flow	-$5.9B	First negative	Pressured

The moat

Google Cloud hypergrowth: +82% YoY revenue, tripled operating income, $514B contracted backlog converting over 24 months.

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Alphabet Inc. valuation framework

The growth engine

Gemini scale: 950M MAU, 22B API tokens per minute, deepening enterprise AI adoption on Google infrastructure.

Three scenarios make the valuation range explicit

Three scenarios, weighted 25% / 50% / 25% into the blended target.

SCENARIO	WEIGHT	2027-31 CAGR	2031 REVENUE	2031 EBIT MARGIN	TARGET	UPSIDE
Bear	25%	8.0%	$719.97B	33.5%	**$78.49**	-76%
Base	50%	12.0%	$863.55B	36.5%	**$149.97**	-53%
Bull	25%	16.0%	$1,029.17B	40.0%	**$265.38**	-18%

Bear: downside case at $78.49 (-76%).

Base: central case at $149.97 (-53%).

Bull: upside case at $265.38 (-18%).

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The probability-blended target implies 50% downside

A 25% Bear / 50% Base / 25% Bull weighting implies -50% from the $322.48 reference price.

$160.95
BLENDED TARGET

The valuation uses a multi-year unlevered free cash flow DCF. The discount rate is the live CAPM rate: the 10-year Treasury yield plus beta times the market-implied equity risk premium. Terminal value is a Gordon perpetuity on terminal-year free cash flow.

Mechanics are explicit. The bridge runs revenue to EBIT to cash taxes, then adds back D&A and subtracts capex and working-capital investment to reach unlevered free cash flow. Targets are divided by diluted shares outstanding.

Valuation horizon	**2027-2031**
Live WACC	**9.37%**
WACC path (Year 1 to final year)	**9.37% to 8.78%**
Terminal growth	**2.0% / 2.5% / 3.0%**
Net cash	**$142.31B**
2031 FCF (Base)	**$148.43B**
10-year Treasury yield	**5.0%**
Beta	**1.21**
Equity risk premium	**3.7%**

HOLD, VALUATION CONCERNS | $160.95 BLENDED TARGET | WEN

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The reverse DCF: what the market prices in

Holding the base-case FCF margin flat, the market-implied revenue CAGR is 59.4%, at/above Bull (16%) - euphoric.

Reverse DCF sensitivity

Implied enterprise value and share price at each five-year revenue CAGR, with the base-case free cash flow margin held flat. The highlighted row sits nearest the market-implied CAGR.

CAGR	Implied EV	Implied price	CAGR	Implied EV	Implied price	CAGR	Implied EV	Implied price
-20%	$170.8B	$25.43	+0%	$438.7B	$47.20	+20%	$1003.0B	$93.05
-19%	$179.5B	$26.14	+1%	$458.5B	$48.81	+21%	$1042.4B	$96.24
-18%	$188.6B	$26.89	+2%	$479.0B	$50.48	+22%	$1083.0B	$99.55
-17%	$198.2B	$27.66	+3%	$500.3B	$52.21	+23%	$1124.9B	$102.95
-16%	$208.2B	$28.47	+4%	$522.4B	$54.00	+24%	$1168.2B	$106.47
-15%	$218.6B	$29.32	+5%	$545.2B	$55.86	+25%	$1212.8B	$110.09
-14%	$229.5B	$30.20	+6%	$569.0B	$57.78	+26%	$1258.8B	$113.83
-13%	$240.8B	$31.12	+7%	$593.5B	$59.78	+27%	$1306.3B	$117.68
-12%	$252.6B	$32.09	+8%	$619.0B	$61.85	+28%	$1355.2B	$121.66
-11%	$265.0B	$33.09	+9%	$645.4B	$63.99	+29%	$1405.6B	$125.75
-10%	$277.8B	$34.13	+10%	$672.7B	$66.21	+30%	$1457.5B	$129.97
-9%	$291.2B	$35.22	+11%	$701.0B	$68.51	+31%	$1511.0B	$134.32
-8%	$305.1B	$36.35	+12%	$730.2B	$70.89	+32%	$1566.1B	$138.79
-7%	$319.6B	$37.53	+13%	$760.5B	$73.35	+33%	$1622.8B	$143.40
-6%	$334.7B	$38.75	+14%	$791.8B	$75.89	+34%	$1681.2B	$148.15
-5%	$350.4B	$40.03	+15%	$824.2B	$78.52	+35%	$1741.3B	$153.03
-4%	$366.8B	$41.36	+16%	$857.7B	$81.24	+36%	$1803.2B	$158.06
-3%	$383.7B	$42.74	+17%	$892.2B	$84.05	+37%	$1866.9B	$163.23
-2%	$401.4B	$44.17	+18%	$928.0B	$86.95	**+38%**	**$1932.3B**	**$168.55**
-1%	$419.7B	$45.66	+19%	$964.9B	$89.95			

Balance sheet and share count reconciliation

Enterprise value converts to equity value after net debt, and targets divide by diluted shares.

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SCENARIO	ENTERPRISE VALUE	EXIT EV/ FCF	TERMINAL GROWTH	PLUS: NET CASH	EQUITY VALUE	TARGET
Bear	$823.76B	—	2.0%	$142.31B	$966.07B	**$78.49**
Base	$1,703.67B	—	2.5%	$142.31B	$1,845.98B	**$149.97**
Bull	$3,124.30B	—	3.0%	$142.31B	$3,266.61B	**$265.38**

Terminal value share and terminal WACC

We disclose how much of each scenario value sits in the terminal year. Terminal value is 92.1% of Base enterprise value (Bear 95.6%, Bull 89.6%). The terminal WACC is 8.78% in Base (Bear 8.78%, Bull 8.78%), on a terminal beta of 1.05 (Bear 1.05, Bull 1.05).

Fair value across scenarios

Model fair value by year under each scenario, on the same paths the ticker page chart draws.

SCENARIO	2026	2027	2028	2029	2030	2031
Bear	$78.49	$85.84	$96.92	$108.04	$117.94	$124.95
Base	$149.97	$164.02	$182.54	$200.76	$216.91	$229.85
Bull	$265.38	$290.25	$318.51	$346.56	$372.30	$394.00

Methodology note. The discount rate is the live U.S. 10-year Treasury yield plus beta times equity risk premium. Terminal growth reflects long-run nominal GDP. The $160.95 headline target applies 25% / 50% / 25% Bear / Base / Bull weights.

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The franchise deep dive

Google Cloud hypergrowth: +82% YoY revenue, tripled operating income, $514B contracted backlog converting over 24 months.

Gemini scale: 950M MAU, 22B API tokens per minute, deepening enterprise AI adoption on Google infrastructure.

TPU monetization: first external sales of complete TPU systems, a new hardware revenue stream from 2027.

Capital return: Berkshire Hathaway's $10B endorsement; ongoing buybacks steadily reduce share count (12.9B to 12.3B diluted since the split).

Revenue and margin trend

METRIC	FY2022	FY2023	FY2024	FY2025	H1 2026
Revenue ($B)	282.8	307.4	350.0	402.8	229.7
Operating income ($B)	74.8	84.3	112.4	129.0	80.5
Operating margin	26.5%	27.4%	32.1%	32.0%	35.0%
Free cash flow ($B)	60.0	69.5	72.8	73.3	N/A
Capex ($B)	31.5	32.3	52.5	91.4	80.6

Valuation cross-check

DCF base **$149.97** vs $322.48 (-53%). Blended (25/50/25) **$160.95**. Even the $265.38 Bull case is 18% below the price. The discount path runs from the live CAPM WACC to a terminal WACC of 8.8% on the decayed terminal beta of 1.05, with 2.5% terminal growth in the Base case.

Reverse DCF basis: we hold the Base bridge's five-year average FCF margin flat (6.1%, an average that spans the 2027-28 capex trough, when Base FCF is negative, rather than the 17.2% terminal-year margin) and discount at the Base terminal WACC. On that trough-average basis the market price implies a five-year revenue CAGR far above every scenario; on a terminal-year margin basis the implied growth rate would be materially lower. We disclose the basis because the choice drives the headline implied CAGR.

At ~8.5x FY2026E revenue and 22.8x forward earnings, GOOGL trades at a premium to slower-growing mega-caps but in line with AI-infrastructure peers. The premium is unsupported by near-term FCF, which is negative under the capex load.

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Historically GOOGL traded at 18-25x earnings with 20%+ FCF margins. The current multiple embeds flawless AI-capex ROI; any guide-down in Cloud growth or capex discipline re-rates the multiple lower, as the September 3.8% selloff previewed.

What to monitor

~Oct 29, 2026. Q3 2026 earnings: watch Search and Cloud growth, capex run-rate, and FCF trajectory.

Q4 2026. 2027 capex budget signal: a below-$200B guide with Cloud margins intact would be the single biggest positive re-rating event.

2027. TPU system sales ramp to external customers, a new hardware revenue stream.

Ongoing. Waymo city expansion and international launches following the $16B round at a $126B valuation.

Financial summary: income statement

YEAR	REVENUE	GROSS PROFIT	OPERATING INCOME	EBITDA
2021	$257.6B	—	$78.7B	—
2022	$282.8B	$156.6B	$74.8B	$85.2B
2023	$307.4B	$174.1B	$84.3B	$98.0B
2024	$350.0B	$203.7B	$112.4B	$135.4B
2025	$402.8B	$240.3B	$129.0B	$180.7B

Balance sheet

YEAR	TOTAL ASSETS	TOTAL LIABILITIES	TOTAL EQUITY	CASH
2021	$359.3B	$107.6B	$251.6B	$20.9B
2022	$365.3B	—	$256.1B	$21.9B
2023	$402.4B	—	$283.4B	$24.0B
2024	$450.3B	—	$325.1B	$23.5B
2025	$595.3B	—	$415.3B	$30.7B

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Cash flow

YEAR	OPERATING CF	CAPEX	FINANCING CF	FREE CF
2021	$91.7B	$-24.6B	$-61.4B	$67.0B
2022	$91.5B	$-31.5B	$-69.8B	$60.0B
2023	$101.7B	$-32.3B	$-72.1B	$69.5B
2024	$125.3B	$-52.5B	$-79.7B	$72.8B
2025	$164.7B	$-91.4B	$-37.4B	$73.3B

Methodology note. The discount rate is the live U.S. 10-year Treasury yield plus beta times equity risk premium. Terminal growth reflects long-run nominal GDP. The $160.95 headline target applies 25% / 50% / 25% Bear / Base / Bull weights.

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Agency risk score 52 of 100: balanced, typical corporate alignment

The score follows the website's Executive Alignment & Agency Risk model: insider ownership up to 40 points, open-market Form 4 flow over the trailing six months up to 30, and pay-for-performance up to 30. Insiders hold 1.60% of shares outstanding. Form 4 filings over the trailing six months show $0 of open-market purchases against $0 of sales, a net flow of $0. One-year total shareholder return is +32.3% against CEO total pay of $10.91M (Mr. Sundar Pichai), which the model scores as PERFORMANCE EARNED.

52
AGENCY RISK SCORE, OF 100

The three components

COMPONENT	SCORE	WHAT THE SITE SHOWS
Insider Ownership (Skin in the Game)	8 / 40	1.60% of shares held by insiders
Form 4 Flow (Conviction Flow)	18 / 30	Buys $0 (0) vs sales $0 (0), net $0, trailing six months
Pay-for-Performance	26 / 30	PERFORMANCE EARNED: Stock Outperformance Matches Comp Structure. 1Y TSR +32.3%; Mr. Sundar Pichai, CEO & Director, total pay $10.91M
Agency Risk Score	**52 / 100**	Regime: BALANCED / TYPICAL CORPORATE

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SEC proxy audit

N/A. The weekly DEF 14A proxy pipeline holds no CD&A audit record for this ticker, so no audit findings are shown.

Methodology. This module reuses the website's Executive Alignment & Agency Risk model unchanged. Scores of 75 and above band as founder-mode, high conviction; 45 to 74 as balanced, typical corporate; below 45 as high agency risk. Inputs are the same feeds behind the site model (Yahoo Finance insider holdings, Form 4 transactions and executive compensation) plus the SEC DEF 14A CD&A audit from the weekly proxy pipeline. Computed 22 October 2026 for this note.

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The bear case

AI capex ROI shortfall

$195-205B of 2026 capex (up ~119% YoY), rising again in 2027, with Q2 FCF already negative. If Cloud and AI revenue do not compound fast enough, shareholders fund a multi-year build with no return.

Search disruption

Meta's Muse agent bypasses the search results page where Google's ad auction operates; 2.8M downloads in twelve days triggered a 3.8% selloff. Agentic AI is the first structural threat to Search monetization.

Regulatory remedies

Ad-tech rulings require business changes and the search case forces data sharing with competitors plus limits on exclusive distribution deals, chipping at structural advantages.

Valuation

The market prices a ~59% five-year revenue CAGR per our reverse DCF. Any growth disappointment against that bar implies sharp multiple compression.

YouTube competition

Q2 YouTube ad revenue missed estimates on competition and shifting marketing budgets, removing a cushion that usually offsets Cloud investment pressure.

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The strengths

Operating Quality

One of the highest-quality businesses in the market: 32-36% operating margins, $129B FY2025 operating income, and a fortress balance sheet ($100.2B of debt against $242.5B of cash plus current marketable securities, a net cash position of $142.3B on the house definition). ROIC remains elite on the ad franchise.

Recent Trend

Revenue accelerating: Q1 2026 +21.8%, Q2 2026 +24.4% YoY; Google Cloud +82% YoY; operating margin expanded to 34-36%. Offset by Q2 free cash flow turning negative ($5.9B) as capex doubled, and the stock's 3.8% September selloff on AI-agent disruption fears.

Forward Growth

Google Cloud carries a $514B backlog (nearly doubled in twelve months) with tripled operating income; Gemini app at 950M MAU processing 22B tokens per minute; TPU system sales to external customers beginning; Waymo valued at $126B after a $16B round. Multiple compounding growth engines.

Valuation

Our DCF base target is $149.97 vs $322.48 (-53%); even the Bull case ($265.38) sits 18% below the price. The reverse DCF shows the market pricing a ~59% five-year revenue CAGR. At 8.5x FY2026E revenue with FCF under acute capex pressure, the multiple discounts perfection.

Cascadia scorecard

Composite score 64: Fair. Confidence High, as of 2026-10-18.

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PILLAR	RATIONALE	SCORE
Operating Quality	One of the highest-quality businesses in the market: 32-36% operating margins, $129B FY2025 operating income, and a fortress balance sheet ($100.2B of debt against $242.5B of cash plus current marketable securities, a net cash position of $142.3B on the house definition). ROIC remains elite on the ad franchise.	88
Recent Trend	Revenue accelerating: Q1 2026 +21.8%, Q2 2026 +24.4% YoY; Google Cloud +82% YoY; operating margin expanded to 34-36%. Offset by Q2 free cash flow turning negative ($5.9B) as capex doubled, and the stock's 3.8% September selloff on AI-agent disruption fears.	72
Forward Growth	Google Cloud carries a $514B backlog (nearly doubled in twelve months) with tripled operating income; Gemini app at 950M MAU processing 22B tokens per minute; TPU system sales to external customers beginning; Waymo valued at $126B after a $16B round. Multiple compounding growth engines.	85
Valuation	Our DCF base target is $149.97 vs $322.48 (-53%); even the Bull case ($265.38) sits 18% below the price. The reverse DCF shows the market pricing a ~59% five-year revenue CAGR. At 8.5x FY2026E revenue with FCF under acute capex pressure, the multiple discounts perfection.	25
Catalysts & Risk	Near-term catalysts: Q3 2026 earnings (~Oct 29), 2027 capex budget signal, Waymo expansion. But the risk ledger is heavy: a $200B+ annual capex supercycle with uncertain returns, Meta's Muse agent bypassing the search results page, ad-tech and search antitrust remedies, and YouTube share pressure.	55

Investment view

A credible 2027 capex normalization (below $200B with Cloud margins holding), Search monetization re-accelerating past AI-agent fears, or a 10Y yield decline cutting our ~8.9% WACC would lift the DCF materially. Conversely, another capex guide raise, Cloud deceleration, or Search share loss would cut it further.

Investment view: Hold, Valuation Concerns. The $160.95 blended target implies -50%, balancing the $78.49 / $149.97 / $265.38 Bear / Base / Bull outcomes.

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